

PROJECT REPORT- DIGITAL APPROVAL SYSTEM

Course name: **OBJECT ORIENTED
PROGRAMMING**

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Abstract

The Paper Approval System is a Java Swing-based desktop application designed to simplify and digitize the process of academic paper approval within a college environment.

This system allows students to submit papers for approval, and faculty members (such as tutors, HODs, principals, and deans) to review and approve them systematically.

It minimizes manual paperwork, ensures transparency, and helps track approval status in real-time.

Objectives

- To digitize and simplify the academic paper approval process.
- To enable easy registration and login for both students and faculty members.
- To track the paper status at different stages (Tutor → HOD → Principal → Dean).
- To maintain a secure and role-based login system.
- To provide a simple and beginner-friendly GUI using Java Swing.

Scope

- The system covers all departments and their respective classes:
- Departments: CSE, EEE, ECE, CIVIL, MECH
- Each department has multiple classes and respective tutors.
- CSE has 6 first-year classes, 4 second-year classes, 3 each in third and fourth years.
- Other departments have 4 classes (one per year).
- Faculty and students are registered separately.

Modules of the System

(a) User Module

- Handles registration and login for both students and faculty.
- Ensures role-based access — student/faculty accounts are separated.

(b) Paper Submission Module

- Students can upload or submit their papers for approval.
- Displays the status (“Waiting for Tutor Approval”, “Approved by Tutor”, etc.).
- Shows tutor name and confirmation pop-ups.

(c) Approval Module

- Faculty can view papers submitted by students of their class/department.
- Tutors → HOD → Principal → Dean approval workflow.
- Each approval updates the paper status.

(d) Administration Module

- Manages the list of departments, tutors, and registered users.
- Faculty accounts are pre-registered with simple usernames and a common password (e.g., faculty)

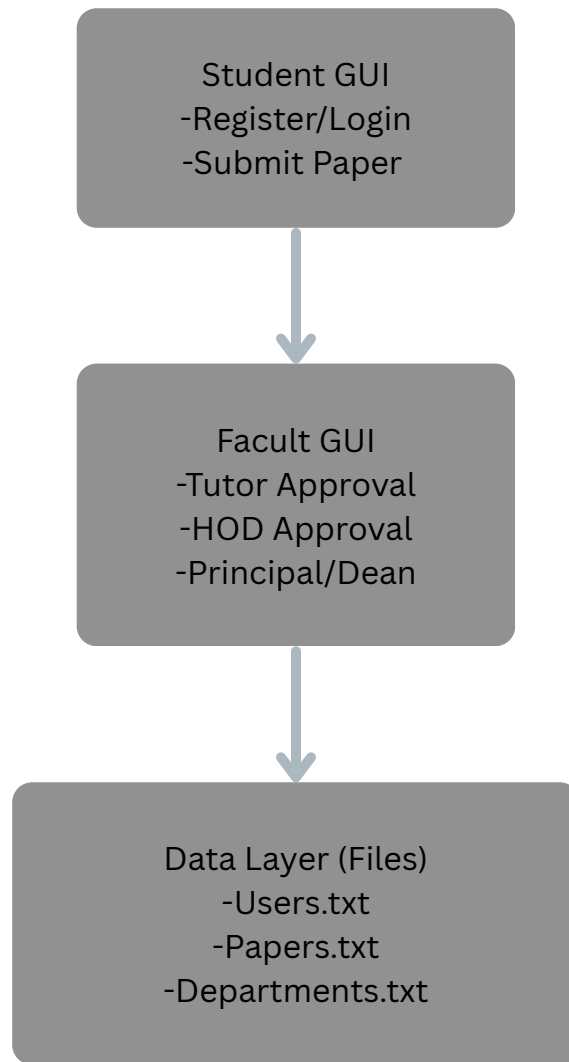
Technologies Used

Technology	Purpose
Java (JDK 17+)	Core programming language
Java Swing (JFrame, JPanel, JButton, etc.)	GUI Design
AWT	Basic UI Components
File Handling (Java I/O)	Data storage and retrieval
VS Code	IDE used for coding
Linux	Operating system
TXT Files	For lightweight data management

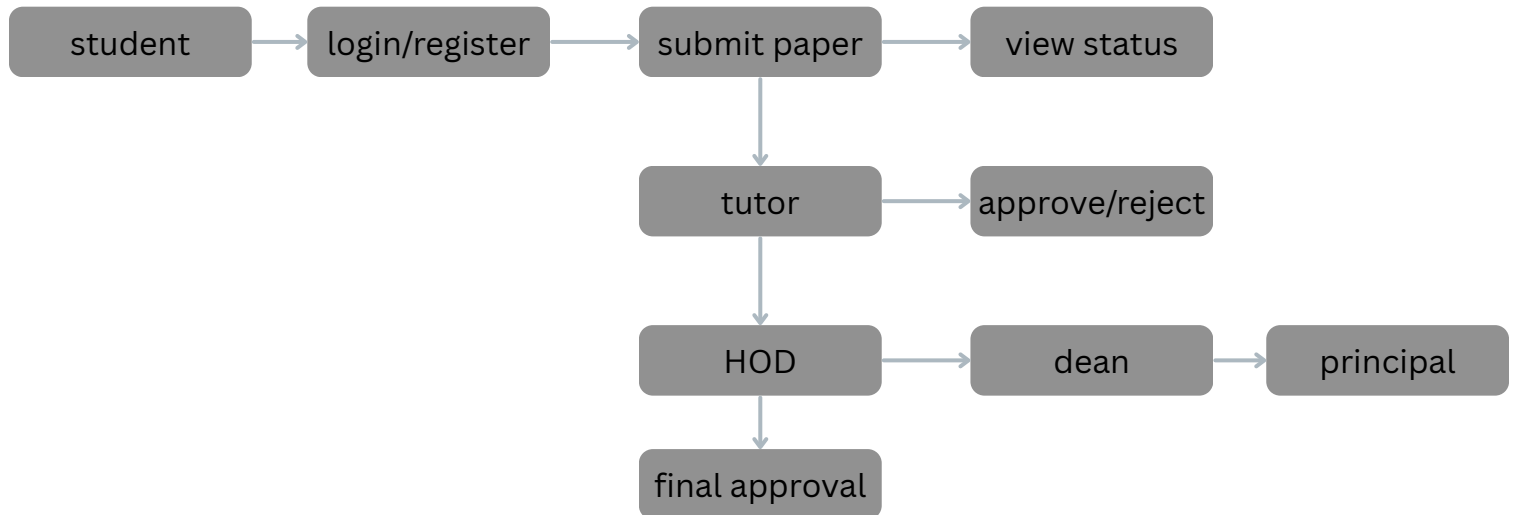
As beginners, we chose beginner-friendly technologies that don't require advanced frameworks or databases like MySQL.

This helped us focus more on learning logic, GUI design, and user interaction.

System Architecture



Functional Diagram



Test Cases

Test Case No.	Description	Input	Expected Output	Result
1	Student registration	New username	Registration successful	Pass
2	Duplicate registration	Existing username	Error message	Pass
3	Student login	Valid credentials	Dashboard opens	Pass
4	Invalid login	Wrong password	Error message	Pass
5	Paper submission	Paper details entered	Status: Waiting for Tutor Approval	Pass
6	Tutor approval	Tutor approves	Status changes to "Approved by Tutor"	Pass
7	Rejection flow	Faculty rejects	Status changes to "Rejected by Tutor"	Pass
8	Final approval	Dean approves	Status: "Fully Approved"	Pass

System Modules

Module	Description
Registration & Login Module	Handles secure login for tutors, HODs, Deans, and Principal. Includes username auto-suggestion and validation.
Paper Submission Module	Allows tutors to upload papers for approval. Each paper is tagged by department and subject.
Approval Module	Allows higher authorities (HOD, Dean, Principal) to review and approve/reject papers. Pending approvals are highlighted.
User Interface Module	Provides dashboards tailored to user roles with relevant functionalities.
Database Management Module	Stores user data, approval status, and submission records. Ensures smooth data flow between modules.
Notification & Status Tracking Module	Keeps users informed about paper status (Approved, Rejected, Pending).

User Manual

A. For Students

Step 1: Registration

1. Open the application.
2. Select “Student Registration.”
3. Fill in details such as:
 - Full Name
 - Department
 - Class (e.g., CSE-A)
 - Username
 - Password
4. Click Register — A confirmation message “Registration Successful!” appears.

Step 2: Login

1. Choose “Student Login.”
2. Enter your username and password.
3. If valid, the main Student Dashboard opens.

Step 3: Submit a Paper

1. On the dashboard, click “Submit Paper.”
2. Enter your paper title and description.
3. Click “Submit.”
4. A pop-up message appears:
5. “Paper sent to [Tutor Name] for approval.”
6. Your paper status updates to “Waiting for Tutor Approval.”

Step 4: View Approval Status

1. Go to “View Paper Status.”
2. The system shows whether your paper is:
 - Waiting for Tutor Approval
 - Approved by Tutor / HOD / Principal / Dean
 - Rejected at any stage

Step 5: Logout

Click the “Logout” button to safely exit your account.

B. For Faculty (Tutor, HOD, Principal, Dean)

Step 1: Login

- Select “Faculty Login.”
- Enter your pre-assigned username and the default password (faculty123).
- If valid, the Faculty Dashboard appears.

Step 2: View Pending Papers

- On the dashboard, click “Pending Papers.”
- A list of papers from students under your supervision appears.

Step 3: Approve or Reject Papers

- Click on a paper to open its details.
- Choose Approve or Reject.
- If approved, the paper moves to the next faculty level.
- Pop-up confirmation:
- “Paper forwarded to [Next Role Name] for approval.”

Step 4: View Approved Papers

Faculty can view all papers they’ve approved for recordkeeping.

Step 5: Logout

Click “Logout” once done to close the session.

C. Error Handling / Notifications

- If invalid credentials are entered, an error message appears.
- If registration uses an existing username, the user is prompted to choose another.
- Pop-up notifications ensure users always know what action occurred (Submitted, Approved, Rejected, etc.).

Development Process

1. Requirement Collection

- Collected the names of all tutors, HODs, Deans, and the Principal from the academic departments.
- Identified workflow levels and responsibilities for each role.

2. User Data Cleaning

- Removed titles (like Dr., Prof.) from names to maintain consistency.
- Converted names into username friendly formats (no spaces, lowercase).

3. Database Design

- Created tables for users, submissions, and approval stages.
- Established relationships between tutor–HOD–Dean–Principal hierarchies.

4. Interface Design

- Developed user-specific dashboards with intuitive navigation.
- Added highlight indicators for unreviewed submissions.

5. Testing and Validation

- Tested each login level to ensure role-based redirection works.
- Verified username search suggestions and approval workflows.

6. Final Integration

- Linked all modules into a unified digital platform.
- Ensured the system supports scalability for new users and departments.

Key Features

1. Automated User Management

- Faculty, HODs, Deans, and the Principal are pre-registered in the system.
- The system automatically generates valid usernames (without spaces or titles like Dr., Prof., etc.).
- When typing a username during login, users get real-time suggestions for similar names, improving speed and accuracy.

2. Role-Based Access

- Each user role (Tutor, HOD, Dean, Principal) has a dedicated dashboard.
- Each dashboard shows only the papers relevant to that user's department or review level.

3. Paper Approval Workflow

- Students submit papers through tutors.
- Tutors review and forward to HODs.
- HODs approve/reject and forward to the Dean.
- The Dean then reviews and sends the paper to the Principal for final approval.

4. Highlighting Pending Approvals

- In each reviewer's dashboard (Tutor, HOD, Dean, Principal), papers that haven't been reviewed yet are highlighted to attract attention and ensure timely action.

5. Automatic Username Validation

- The system ensures usernames have no spaces.
- A clear message is displayed during registration:
- "Usernames with spaces are not accepted."

6. Secure Login and Authentication

- Passwords are stored securely.
- Only verified users can access their respective dashboards.

7. Fully Digital Workflow

- Eliminates physical paperwork.
- Provides complete transparency and accountability at each approval level.

8. Dynamic and Responsive Design

- The interface is user-friendly and works seamlessly across devices.
- Clear navigation for each user role with consistent design and color cues.

Conclusion

The Paper Approval System achieves its objective of simplifying academic paper workflows in a beginner-friendly, functional, and clear manner.

Students and faculty can easily interact through an intuitive GUI that promotes paperless communication.